Division of Cancer Epidemiology & Genetic

DCEG Human Microbiome & Cancer Epidemiology Short Course

Lecture 5: Statistical Analysis

Methods for microbiome data

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NIH NATIONAL CANCER INSTITUTE

Outline

- Quality Control Analysis – Preprocessing, Transformation
- Visualization
- Differential Abundance
- Alpha Diversity
- Beta Diversity
- Measurement error & reproducibility
- Network Analysis
- Data Integration – Multi-omics

Standard workflow for Statistical Analysis of Microbiome

- Define Study Question, Hypotheses & Assumptions
- Retrieve data
- Preprocess
- Explore
- Analyses
- Model
- Report

Define Study Question, Hypotheses & Assumptions

- State the biological question and primary hypothesis explicitly
- Identify outcome of interest (alpha diversity, taxon abundance, community composition, ...)
- Specify key covariates, confounders, and expected effect direction
- Decisions here cascade into every downstream step

Genus-level count matrix (16S amplicon data)

Samples (j = 1, 2, …, N)

Genera (i = 1, 2, …, M)

	S_1	S_2	S_3	⋯	S_n
Bacteroides	142	0	87	⋯	23
Prevotella	0	310	5	⋯	198
Faecalibacterium	56	74	0	⋯	41
⋮	⋮	⋮	⋮	⋱	⋮
Ruminococcus	9	0	133	⋯	7

x_ij = read count of genus i in sample j (non-negative integer; sparse; columns sum to library size)

16S → amplicon sequencing → ASV table
(rows = samples, columns = taxa, entries = counts)

QC & Preprocessing

- Data quality affects the final results
- "***Garbage in, garbage out***": Failing to assess data quality accurately can lead t erroneous interpretations
- The goal of QC is not only to improve quality of data, but also to understand it and its limitations

Common Steps

- **Summarize data**: # features/samples, summary stats, prevalence
- Inspect library sizes (sequencing depth) per sample – Flag outliers
- Filter **low-quality samples** (very low read counts) and negative/mock controls
- Assess and remove **contaminants**
- Data distribution; Abundance visualization
- Collinearity (heatmaps)

QC & Preprocessing

BRITISH ECOLOGICAL SOCIETY | Methods in Ecology and Evolution

Free Access

A protocol for data exploration to avoid common statistical problems

Alain F. Zuur, Elena N. Ieno, Chris S. Elphick

ELSEVIER
Computational and Structural Biotechnology Journal
Volume 21, 2023, Pages 4804-4815

Mini-review

Data pre-processing for analyzing microbiome data – A mini review

Ruwen Zhou [a], Siu Kin Ng [a], Joseph Jao Yiu Sung [a b], Wilson Wen Bin Goh [a c d], Sunny Hei Wong [a b]

Filtering

- Selecting specific taxa or samples from a dataset based on certain criteria.
- Helps to draw the focus of an analysis on particular sets of samples and/or features.

Common criteria

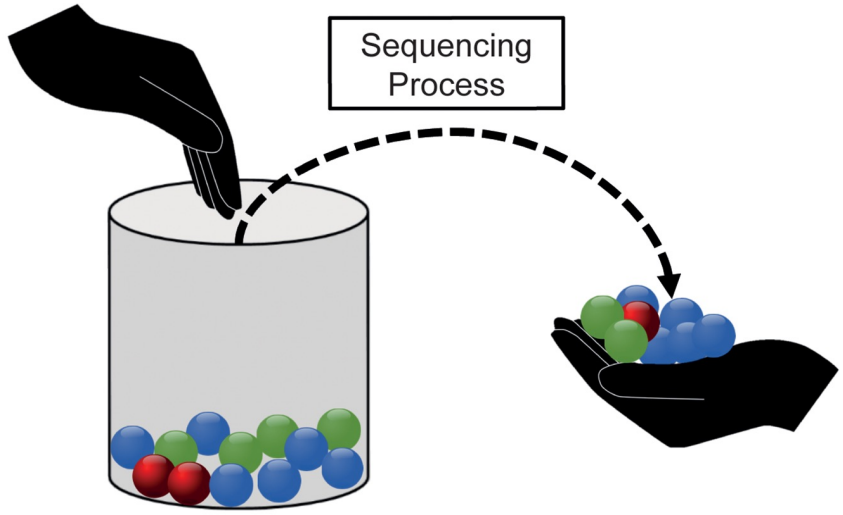
- Subset samples by library size: total number of counts
- Discard features with low variance
- Remove features with very low prevalence (e.g. present in < 5–10% of samples)
- Remove features with near-zero total counts across all samples
- Optionally collapse features to genus level for interpretability and reduced sparsity

Characteristics of Microbiome data

- Compositionality — only relative abundances observed
- Heteroscedasticity — feature abundances vary widely across samples
- Zero-inflation — many taxa absent in most samples

Compositionality

- Sequencing measures **relative abundances**, not true microbial counts
- Values are **parts of a whole**
- Parts are **linked by a fixed total** (library size)
    - If one increases, at least one other must decrease



	True Absolute Abundance	True Relative Abundance	Observed Absolute Abundance	Observed Relative Abundance
	2	1/6	1	1/8
	4	1/3	2	1/4
	6	1/2	5	5/8

Image source: Cappellato, Baruzzo, and Camillo (2022)[12]

Journal of the Royal Statistical Society: Series B (Methodological)

Article

The Statistical Analysis of Compositional Data

J. Aitchison

Microbiome Datasets Are Compositional: And This Is Not Optional

Gregory B. Gloor[1*], Jean M. Macklaim[1], Vera Pawlowsky-Glahn[2] and Juan J. Egozcue[3]

The Microbiome and Epidemiology

It's all relative: analyzing microbiome data as compositions

Gregory B. Gloor PhD[a,*], Jia Rong Wu BSc[a], Vera Pawlowsky-Glahn PhD[b], Juan José Egozcue PhD[c]

[a] Department of Biochemistry, University of Western Ontario, London, Ontario, Canada
[b] Department of Computer Science, Applied Mathematics, and Statistics, University of Girona, Spain
[c] Department of Civil and Environmental Engineering, Technical University of Catalonia, Spain

Analysis of compositions of microbiomes with bias correction

Huang Lin & Shyamal Das Peddada

Nature Communications **11**, Article number: 3514 (2020) | Cite this article

Sparse and Compositionally Robust Inference of Microbial Ecological Networks

Zachary D. Kurtz, Christian L. Müller, Emily R. Miraldi, Dan R. Littman, Martin J. Blaser, Richard A. Bonneau

Inferring Correlation Networks from Genomic Survey Data

Jonathan Friedman, Eric J. Alm

Normalization / Transformation

- Mitigate technical biases in the data.
- Obtain more interpretable values.
- Enhance the comparability of samples/features
- Make data compatible with the assumptions of certain statistical methods.

ELSEVIER

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Relative Abundance (Total Sum Scaling, TSS)

Relative Abundance:

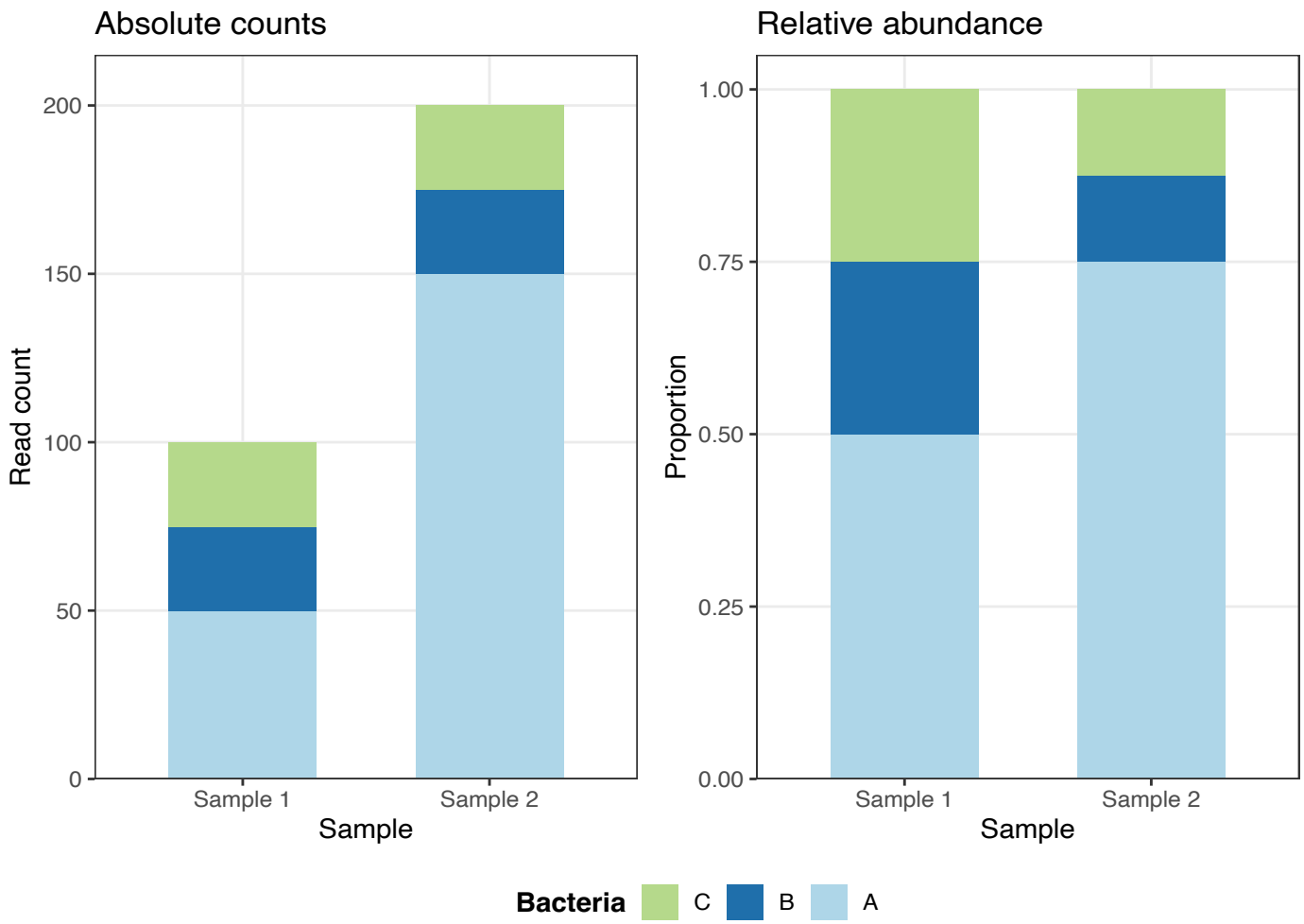
$$X_{ij}^{*} = \frac{Count_{ij}}{\sum_{k=1}^{m} Count_{ik}}$$

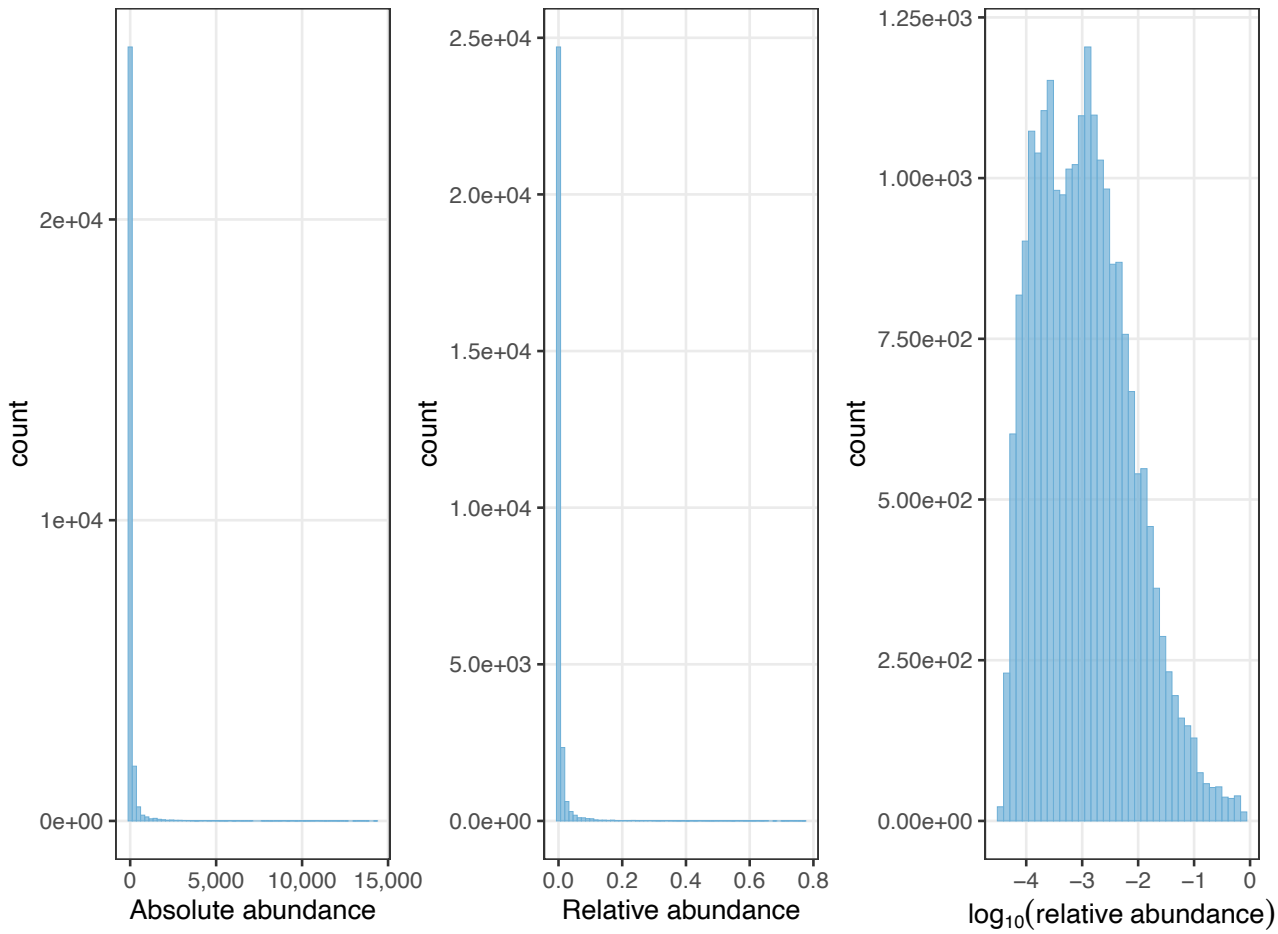
Log-transformed Relative Abundance:

$$X_{ij} = log\left(X_{ij}^{*} + \epsilon\right)$$

- i indexes the sample,
- j indexes the taxon,
- m is the total number of taxa.
- ϵ small pseudo-count

- Removes effect of variable sequencing depth, enabling cross-sample comparison
- Intuitive and widely used — easily interpreted as the fraction of reads attributed to each genus
- Suitable for visualization (bar charts, heatmaps)
- **Limitation**:
  - proportions remain compositional — a true zero cannot be distinguished from a structural zero;
  - log-transformation of proportions requires a pseudo count to handle zeros, which introduces bias


Absolute counts
Relative abundance
Read count
Proportion
200
150
100
50
0
1.00
0.75
0.50
0.25
0.00
Sample 1
Sample 2
Sample
Bacteria
C
B
A


count
2e+04
1e+04
0e+00
0
5,000
10,000
15,000
Absolute abundance
count
2.5e+04
2.0e+04
1.5e+04
1.0e+04
5.0e+03
0.0e+00
0.0
0.2
0.4
0.6
0.8
Relative abundance
count
1.25e+03
1.00e+03
7.50e+02
5.00e+02
2.50e+02
0.00e+00
−4
−3
−2
−1
0
log10(relative abundance)

Centered Log-Ratio (CLR)

- CLR is the principled transformation for compositional data (Aitchison, 1986)
- Centres the data around zero; removes the unit-sum constraint, but with sum-zero constraint.
- Places the data in real (unconstrained) space, enabling use of standard statistical methods

For a sample with genus abundance:

$\mathbf{x} = (x_{1,\ldots,}x_M)$

$$CLR(x_i) = \log\left(\frac{x_i}{g(\mathbf{x})}\right)$$

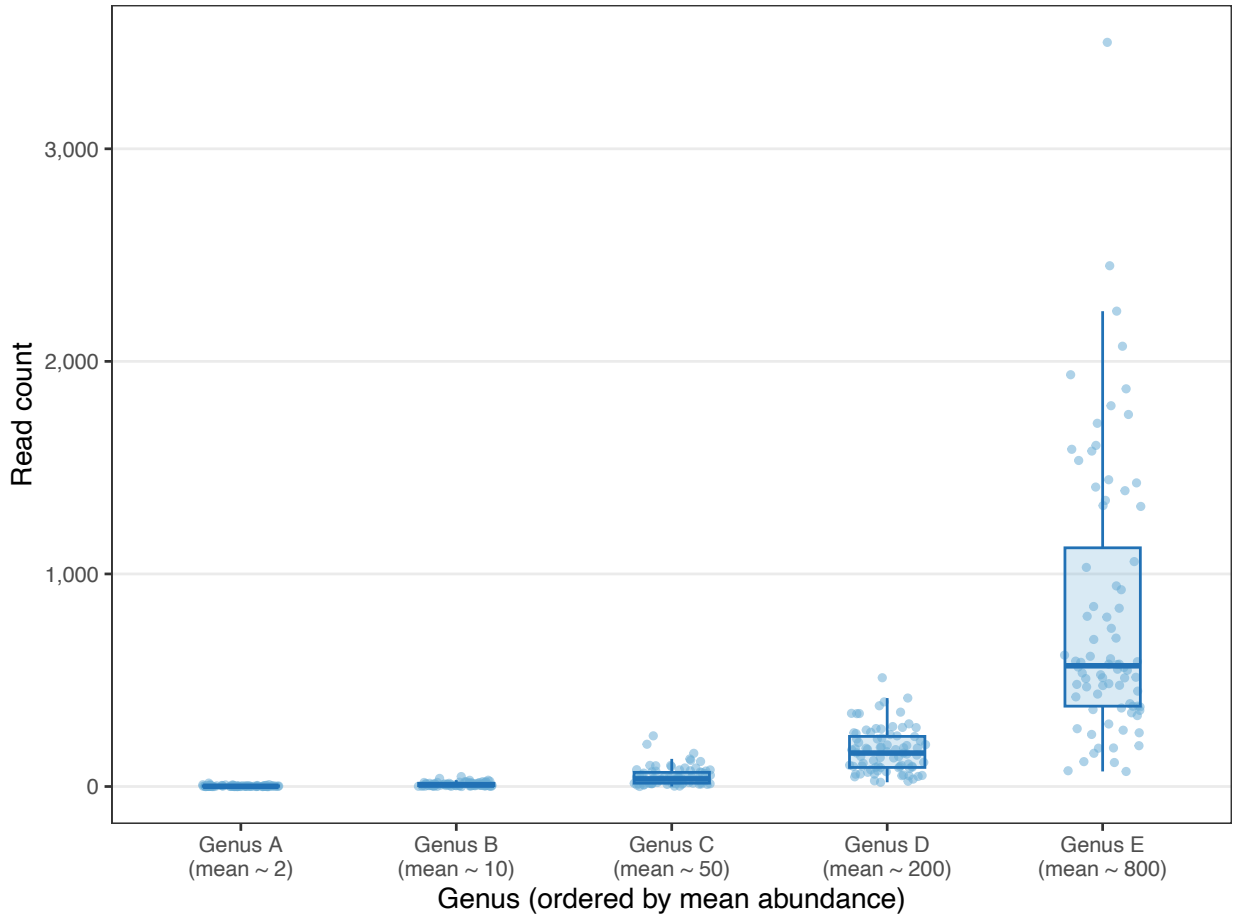
$$= \log(x_i) - \log\big(g(\mathbf{x})\big)$$

$$= \log(x_i) - \frac{1}{M}\sum_{j=1}^{M}\log(x_j)$$

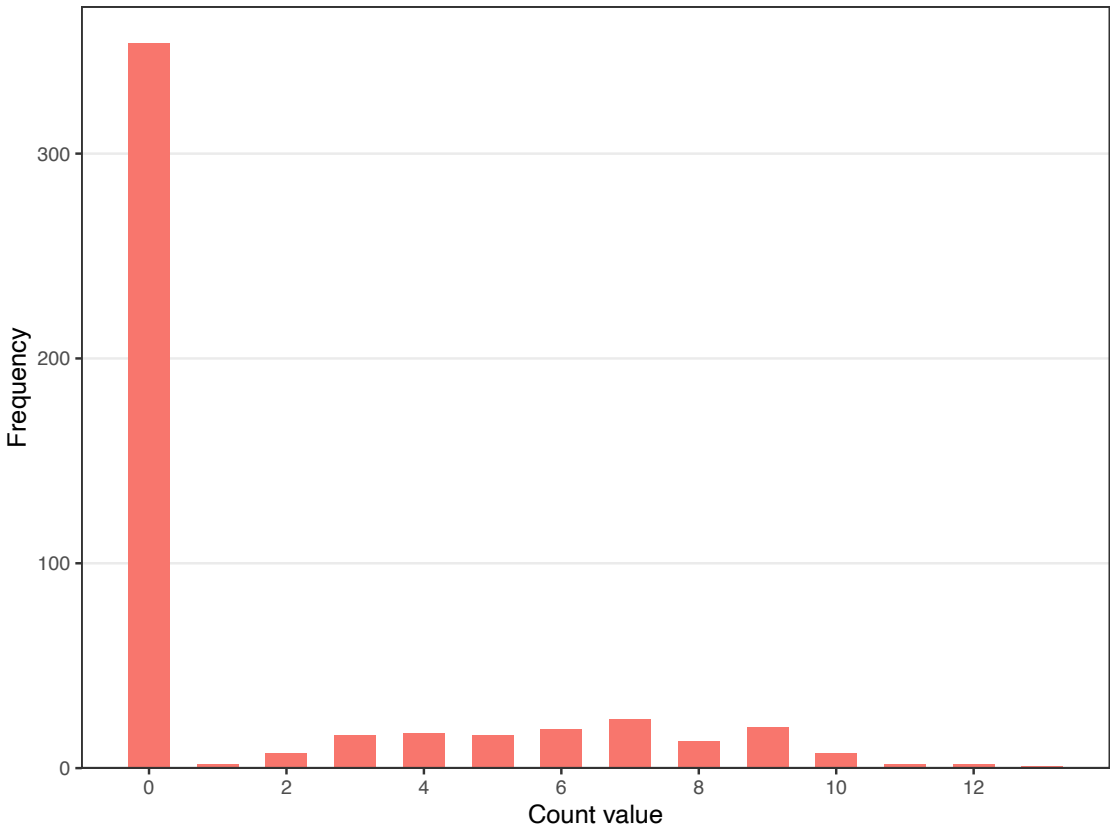
- $g(\mathbf{x}) = \left(\prod_{j=1}^{M} x_j\right)^{1/M}$: geometric mean

- **Key limitation**: requires all values to be strictly positive — zeros must be handled before applying CLR

Heteroscedasticity



Zero Inflation



Other transformation methods (non-exhaustive list)

- **Additive Log-Ratio (ALR):** log-ratio of each genus relative to a single chosen reference genus
- **Hellinger**: square root of relative abundances; reduces the influence of highly abundant taxa
- **Variance-Stabilizing Transformation (VST):** models count overdispersion and stabilizes variance across the abundance range
- **Z-score (standardization):** scales data to zero mean and unit variance.
- **Rank**: replaces abundance values with within-sample ranks
- **Presence/Absence (PA):** binarizes the data — 1 if a genus is detected above a threshold, 0 otherwise
    - Discards all abundance information; appropriate for when detection rather than quantity is the question of interest

- No single transformation is universally optimal.
- The choice should be guided by the research question, the downstream statistical method, and the properties of the data.

Home > Oecologia > Article

Ecologically meaningful transformations for ordination of species data

Published: 01 October 2001

Volume 129, pages 271–280, (2001) Cite this article

Differential Abundance (DA) Analysis

- **Goal**: identify individual taxa whose relative abundances differ between groups (e.g. disease vs. healthy, pre- vs. post-treatment)
- DA findings can point to taxa involved in disease dynamics and generate hypotheses about microbial mechanisms

Method	Type	Key feature
Wilcoxon rank-sum test	Non-parametric	Robust; ignores compositionality
Linear model / t-test (on CLR)	Parametric	Compositionally aware if CLR-transformed
DESeq2	Parametric (NB model)	Widely used; borrowed from RNA-seq
ANCOM-BC2	Compositional	Explicitly corrects for sampling bias
ALDEx2	Compositional (Bayesian)	Propagates uncertainty via Dirichlet sampling
MaAsLin2	Multivariate linear model	Handles covariates, repeated measures

- Apply FDR correction (e.g. Benjamini–Hochberg) to account for multiple testing
- Report effect sizes (log-fold change) alongside p-values

Choosing a DA Method — Practical Guidance

- No single method performs best universally — performance depends on sample size, sparsity, effect size, and normalization choice
- Benchmarking studies have reached discordant conclusions; results are sensitive to data characteristics and pre-processing decisions
- Practical recommendations:
  - For compositional correctness: prefer **ANCOM-BC2** or **ALDEx2**
  - For multivariate/longitudinal designs: use **MaAsLin2**
  - For exploratory or simple two-group comparisons: Wilcoxon on CLR-transformed data is a transparent, defensible baseline

- Run multiple methods and report findings that are robust across methods
- Use the ***benchdamic*** R package (Calgaro et al., 2022) to benchmark and compare DA methods on your own dataset

Alpha Diversity

- **Goal**: quantify the diversity of the microbial community within each sample
  - summarizes richness, evenness, or evolutionary breadth into a single value per sample
- Indices capture different, complementary aspects of within-sample diversity
- Recommended: calculate >=1 index from each category to avoid a partial picture

Category	What it captures	Index	Formula / key idea
Richness	How many distinct ASV/OTUs are present	Observed	Count of genera with count > 0
		Chao1	Corrects for undetected rare taxa using singletons & doubletons
Evenness	Richness and how evenly reads are distributed	Shannon	$-\Sigma\, p_i \log(p_i)$; sensitive to both rare and dominant taxa
		Simpson	Probability that two randomly drawn reads belong to different genera; emphasizes dominant taxa
Phylogenetic	Sum of length of all branches	Faith's PD	Total branch length of the phylogenetic tree spanned by observed taxa

Alpha Diversity — Statistical Comparison Across Groups

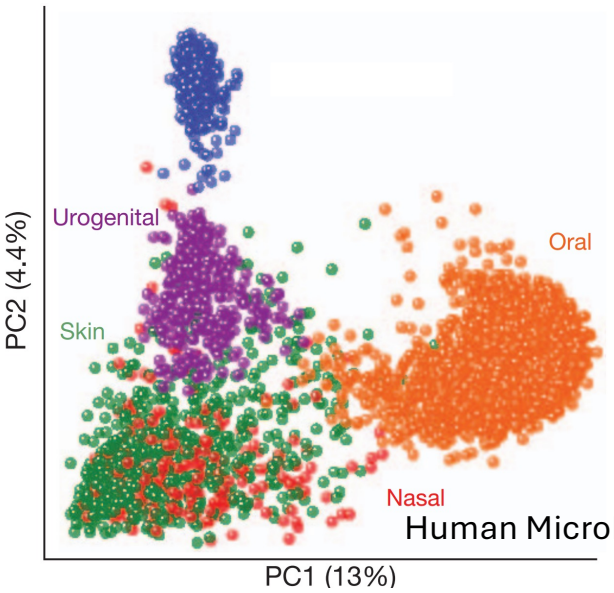
- After computing indices, we ask: does alpha diversity differ between groups?
- Each index produces one value per sample → standard univariate statistical tests

Design	Recommended approach
Two groups, normal distribution	Independent samples t-test or linear model
Two groups, non-normal	Wilcoxon rank-sum test
More than two groups	ANOVA or Kruskal–Wallis + post-hoc tests
Continuous exposure or covariates	Linear regression / ANCOVA
Longitudinal / repeated measures	Linear mixed-effects model

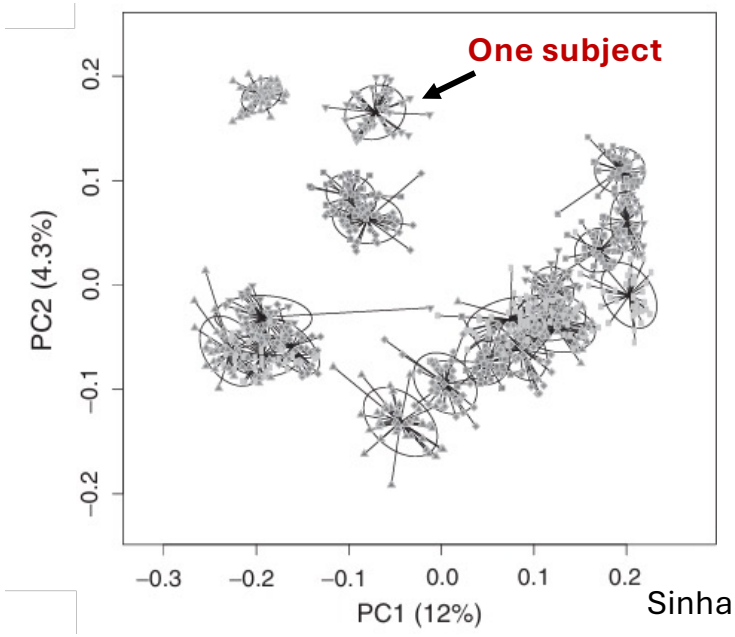
- Always check **distributional assumptions**
- **Adjust for confounders** (age, sex, BMI)
- Visualize with box plots or violin plots
- Interpret with caution: a statistically significant difference in alpha diversity does not identify which taxa drive the difference

Beta-Diversity Analysis

- **Beta-diversity *distance* matrix**
  - Unweighted UniFrac, weighted UniFrac, Bray-Curtis
  - $N \times N$ matrix for N subjects
- **Principal coordinate analysis (PCoA)**



Human Microbiome Project



Sinha et al., 2016

Beta-Diversity Analysis

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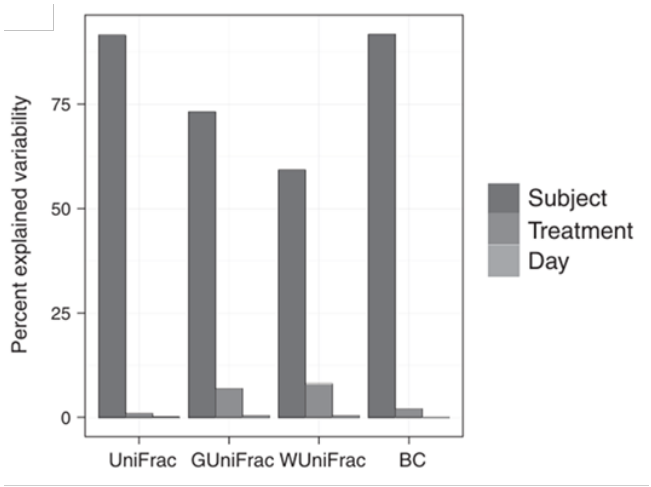
	Meta-analysis Cases = 1306/ subcohort = 3304			AHS[a] Cases = 244/ subcohort = 1010		NIH-AARP[b] Cases = 371/ subcohort = 1077		PLCO[c] Cases = 691/ subcohort = 1217	
Microbiome metrics	HR (95% CI)	P	$P_{heterogeneity}$	HR (95% CI)	P	HR (95% CI)	P	HR (95% CI)	P
PCA vectors from relative abundance[d]									
PC1 (44.8%)	0.99 (0.91 to 1.07)	.72	.93	0.98 (0.83 to 1.17)	.86	1.01 (0.87 to 1.17)	.91	0.97 (0.87 to 1.09)	.64
PC2 (15.2%)	1.11 (1.04 to 1.18)	.003	.43	1.03 (0.90 to 1.18)	.64	1.10 (0.98 to 1.24)	.11	1.15 (1.04 to 1.27)	.005
PCA vectors from presence/absence[d]									
PC1 (7.9%)	1.17 (1.08 to 1.25)	<.001	.92	1.14 (0.98 to 1.33)	.09	1.16 (1.02 to 1.32)	.03	1.18 (1.06 to 1.32)	.003

Vogtmann et al., 2022

Beta-Diversity Analysis

- **Beta-diversity *distance* matrix**
  - Unweighted UniFrac, weighted UniFrac, Bray-Curtis
  - $N \times N$ matrix for N subject
- **Principal coordinate analysis (PCoA)**
- **Distance – based analysis**
  - Full information, most efficient
- **Distance partitioning**
  - R *vegan* package function *adonis*
  - *adonis(Y ~ subjectID + treatment + Day)*
  - *PERANOVA:* two group comparison



McArdle & Anderson, 2001

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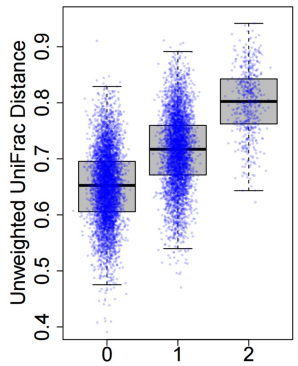
  - R *vegan* package function *adonis*
  - *PERANOVA:* two group comparison
- **Epidemiology study with covariates adjustment**
  - *MiRKAT* (Zhao & Wu, 2015)
  - Convert distance matrix to correlation matrix, linear mixed model

MicrobiomeGWAS: Identify SNPs Associated with Microbiome

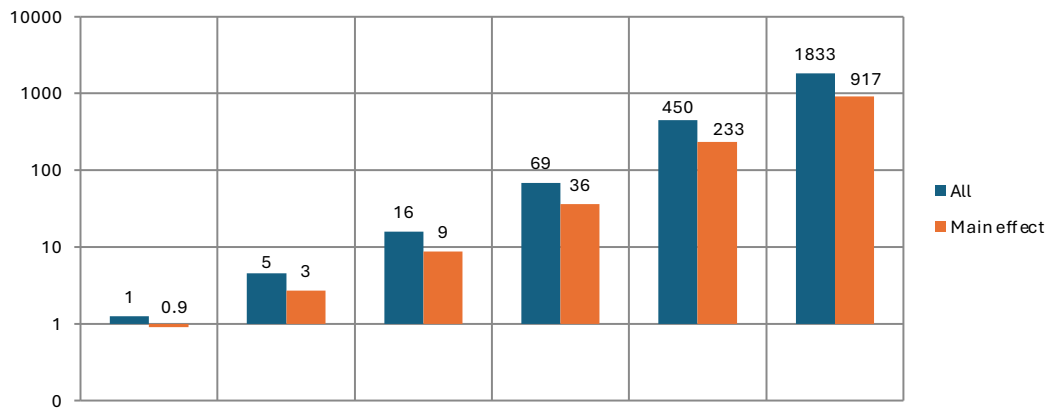
- **Relative abundances & alpha diversity**
  - Linear regression analysis
  - Zero-inflated negative binomial regression
- **Beta-diversity matrix**



$|g_i - g_j|$

Similar genotypes → smaller microbiome distance

Standard approach: adonis-based distance partitioning + permutation
Hundreds of days of computation in a computational cluster.

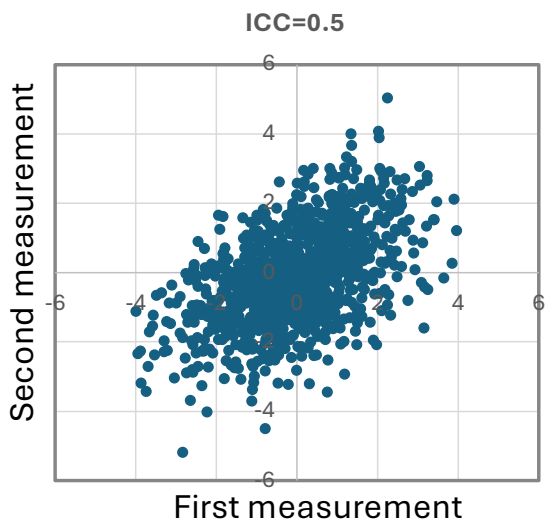


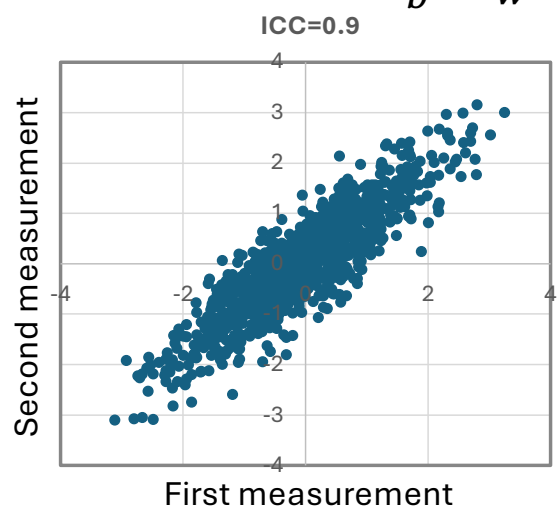
https://github.com/lsncibb/microbiomeGWAS

Hua et al., 2018; Hua et al., 2023

Measurement Error and Implications to Population Studies

- **Measurement error of microbiome metrics**
  - Technical error: sampling, freezing, sequencing, bioinformatics
  - Overtime instability
- **Intraclass correlation coefficient (ICC):** $\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2}$





Measurement Error and Implications to Population Studies

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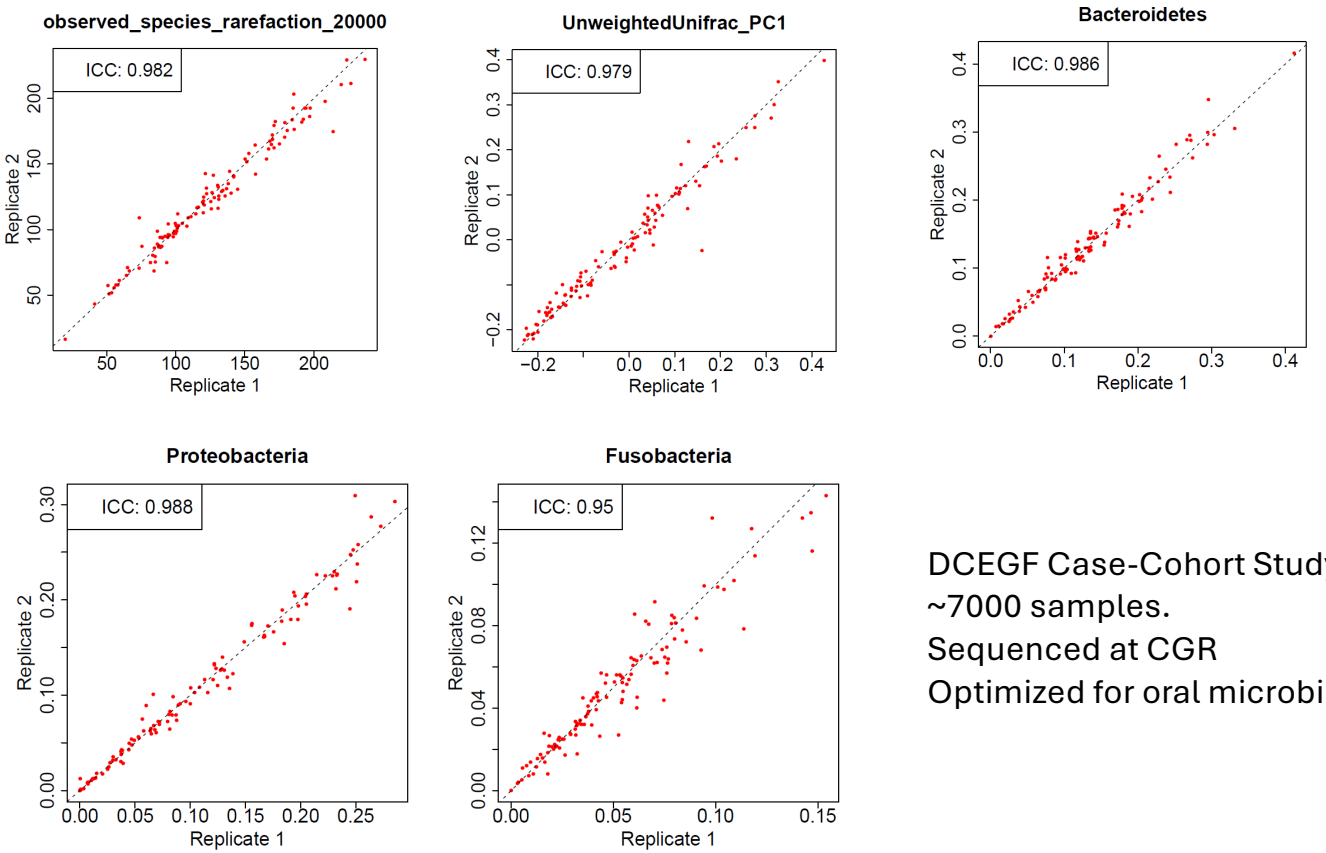
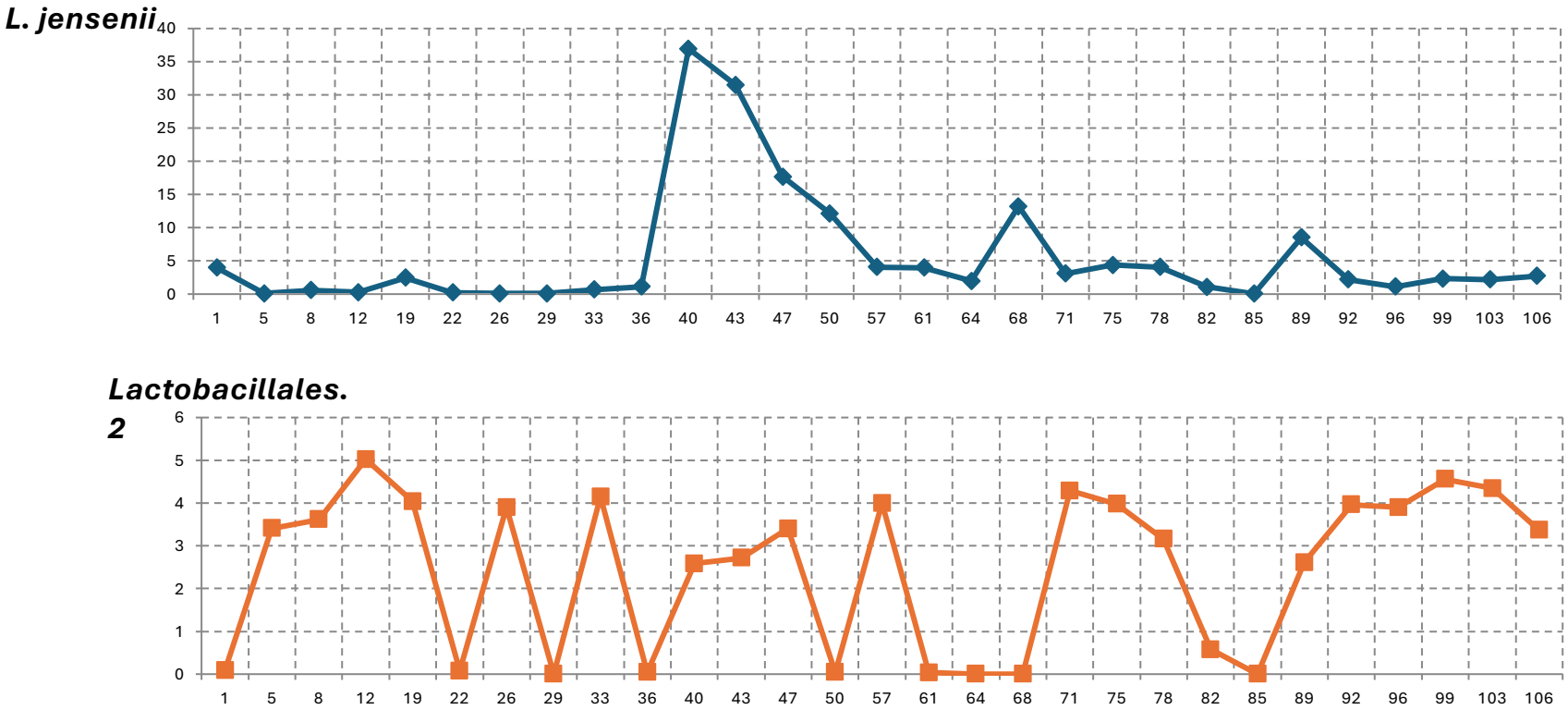
  - Overtime instability
- **Intraclass correlation coefficient (ICC):** $\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2}$
- **Effective sample size loss due to measurement error**
  - N = 1000 and ICC = 30%
  - N = 500 and ICC = 60%
  - N = 333 and ICC = 100%

High Reproducibility in Technical Replicates



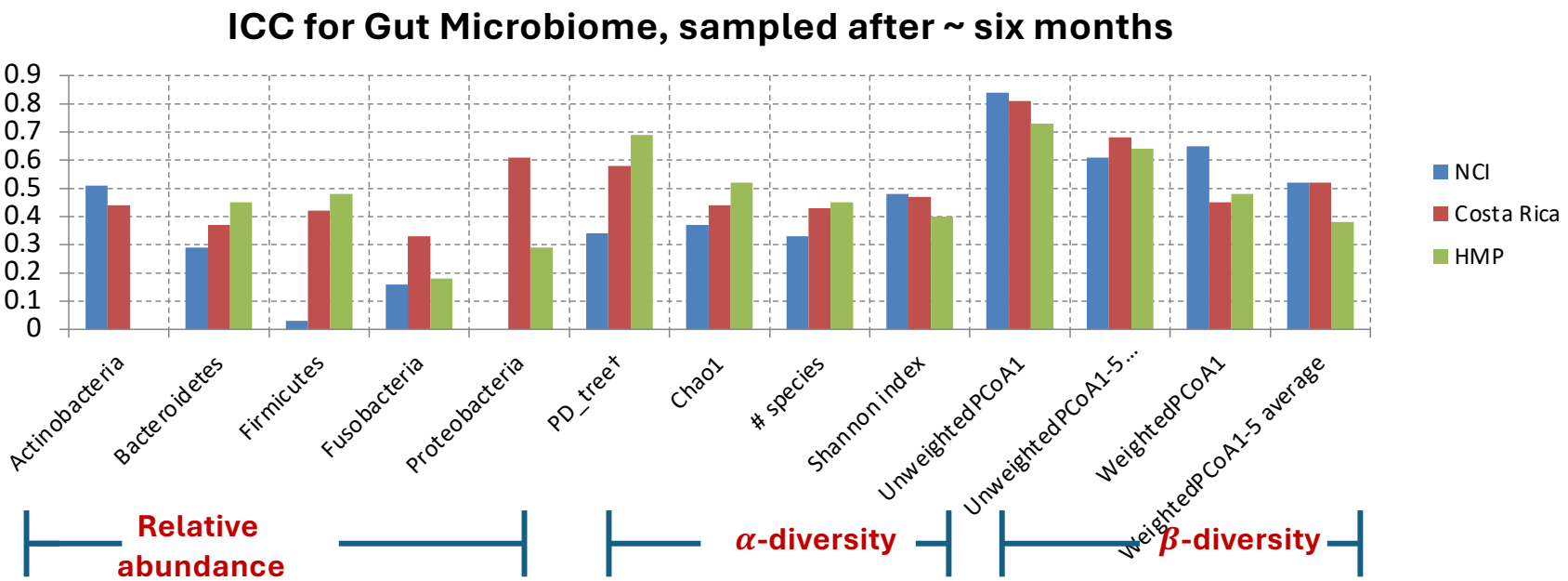
DCEGF Case-Cohort Study
~7000 samples.
Sequenced at CGR
Optimized for oral microbiome

Longitudinal Observation of Vaginal Microbiome



Gajer, et al., Sci. Trans. Med., 2013

ICCs for Microbiome Metrics



Similar results for oral microbiome separated by six months.
Median ICC is 30-35%.

Sinha, ..., Shi, AJE, 2018

Stability of the Fecal and Oral Microbiome over 2 Years at −80°C for Multiple Collection Methods

Semi Zouiouich; Doratha A. Byrd; Xing Hua; Smriti Karwa; Yunhu Wan; Jianxin Shi; Gregory C. Humphrey; Gail L. Ackermann; Rob Knight; Christian C. Abnet; Emily Vogtmann; Rashmi Sinha

Sample Size Estimations Based on Human Microbiome Temporal Stability Over 6 Months: A Shallow Shotgun Metagenome Sequencing Analysis FREE

Semi Zouiouich; Yunhu Wan; Emily Vogtmann; Carolina Porras; Christian C. Abnet; Jianxin Shi; Rashmi Sinha

Comparison of Fecal Sample Collection Methods for Microbial Analysis Embedded within Colorectal Cancer Screening Programs FREE

Comparison of fecal and oral collection methods for studies of the human microbiota in two Iranian cohorts

Zeni Wu[1†], Autumn G. Hullings[1†], Reza Ghanbari[2], Arash Etemadi[1], Yunhu Wan[3,4], Bin Zhu[3,4], Hossein Poustchi[2], Behnam Bagheri Fahraji[5], Mohammad Javad Zare Sakhvidi[6], Jianxin Shi[7], Rob Knight[8,9], Reza Malekzadeh[2*], Rashmi Sinha[1*] and Emily Vogtmann[1]

Comparison of Oral Microbiota Collected Using Multiple Methods and Recommendations for New Epidemiologic Studies

Authors: Yukiko Yano, Xing Hua, Yunhu Wan, Shalabh Suman, Bin Zhu, Casey L. Dagnall, Amy Hutchinson, Kristine Jones, Belynda D. Hicks, Jianxin Shi, Christian C. Abnet, Emily Vogtmann | AUTHORS INFO & AFFILIATIONS

Reproducibility, stability, and accuracy of microbial profiles by fecal sample collection method in three distinct populations

Doratha A. Byrd, Jun Chen, Emily Vogtmann, Autumn Hullings, Se Jin Song, Amnon Amir, Muhammad G. Kibriya, Habibul Ahsan, Yu Chen, Heidi Nelson, Rob Knight, Jianxin Shi, Nicholas Chia, Rashmi Sinha

Comparison of oral collection methods for 16S gene and shotgun metagenomic sequencing

Giovanny Herrera [1], Semi Zouiouich [1], Norma Diaz-Mayoral [2], Vaishna
Britton Trabert [3], Yunhu Wan [1 2], Jia Liu [1 2], Casey L Dagnall [1 2], Kris
Belynda D Hicks [1 2], Amy Hutchinson [1 2], Shilan Li [4], Jianxin Shi [1], Ch
Emily Vogtmann [1]

Quantification of Human Microbiome Stability Over 6 Months: Implications for Epidemiologic Studies

Rashmi Sinha [1], James J Goedert [1], Emily Vogtmann [1], Xing Hua [1], Carolina Porras [2],
rd Hayes [3], Mahboobeh Safaeian [4], Guoqin Yu [1], Joshua Sampson [1], Jiyoung Ahn [3],
n Shi [1]

Analysis | Published: 02 October 2017

Assessment of variation in microbial community amplicon sequencing by the Microbiome Quality Control (MBQC) project consortium

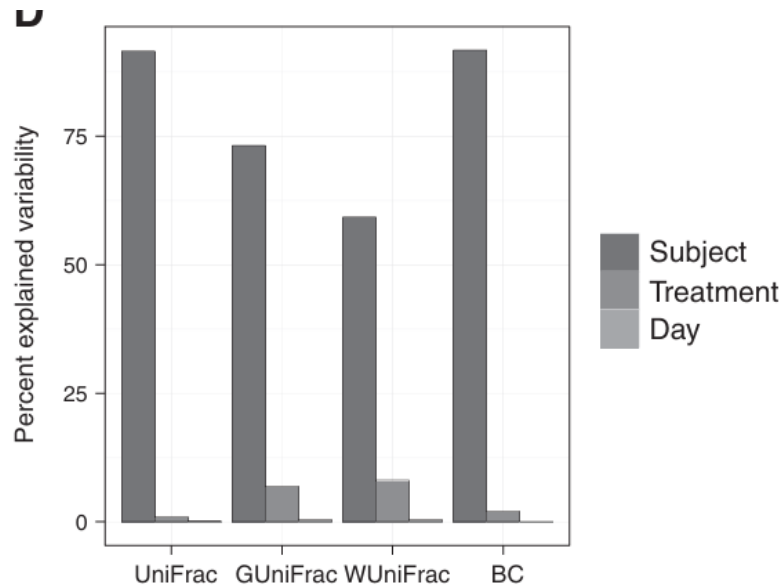
Rashmi Sinha, Galeb Abu-Ali, Emily Vogtmann, Anthony A Fodor, Boyu Ren, Amnon Amir, Emma

Estimate ICC Using Linear Mixed Models

- N subjects with k replicates.
- $y_{ij} = \alpha + b_i + e_{ij}$, where $b_i \sim N(0, \sigma_b^2)$, and $e_{ij} \sim N(0, \sigma_w^2)$.
- $Var(y_{ij}) = \sigma_b^2 + \sigma_e^2$.
- $ICC = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2}$.
- Can adjust for covariates with fixed effects, e.g., age, sex, batches…
- ***ICC depends on transformations.***
- R syntax
  - *library(lme4)*
  - *fit <- lmer(y ~ age + sex + batch + (1 | subject_id), data = dat, REML = TRUE)*

Quantify Measurement Error in Beta-Diversity Matrix

- **Distance-based R2: a motivating example**
  - N subjects, each measured twice. Beta-diversity matrix 2×2.
  - Total distance $D = D_b + D_w$
  - R2 for subjects: $R^2 = D_b/(D_b + D_w)$



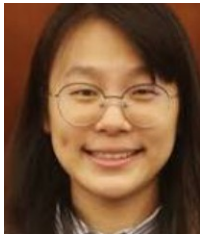
- R *vegan* package function *adonis*
- *adonis(Y ~ subjectID + treatment + Day)*
- Days: soon,1 and 4 days
- Treatment: no, RNA*Later*, ethanol

Sinha et al., 2016

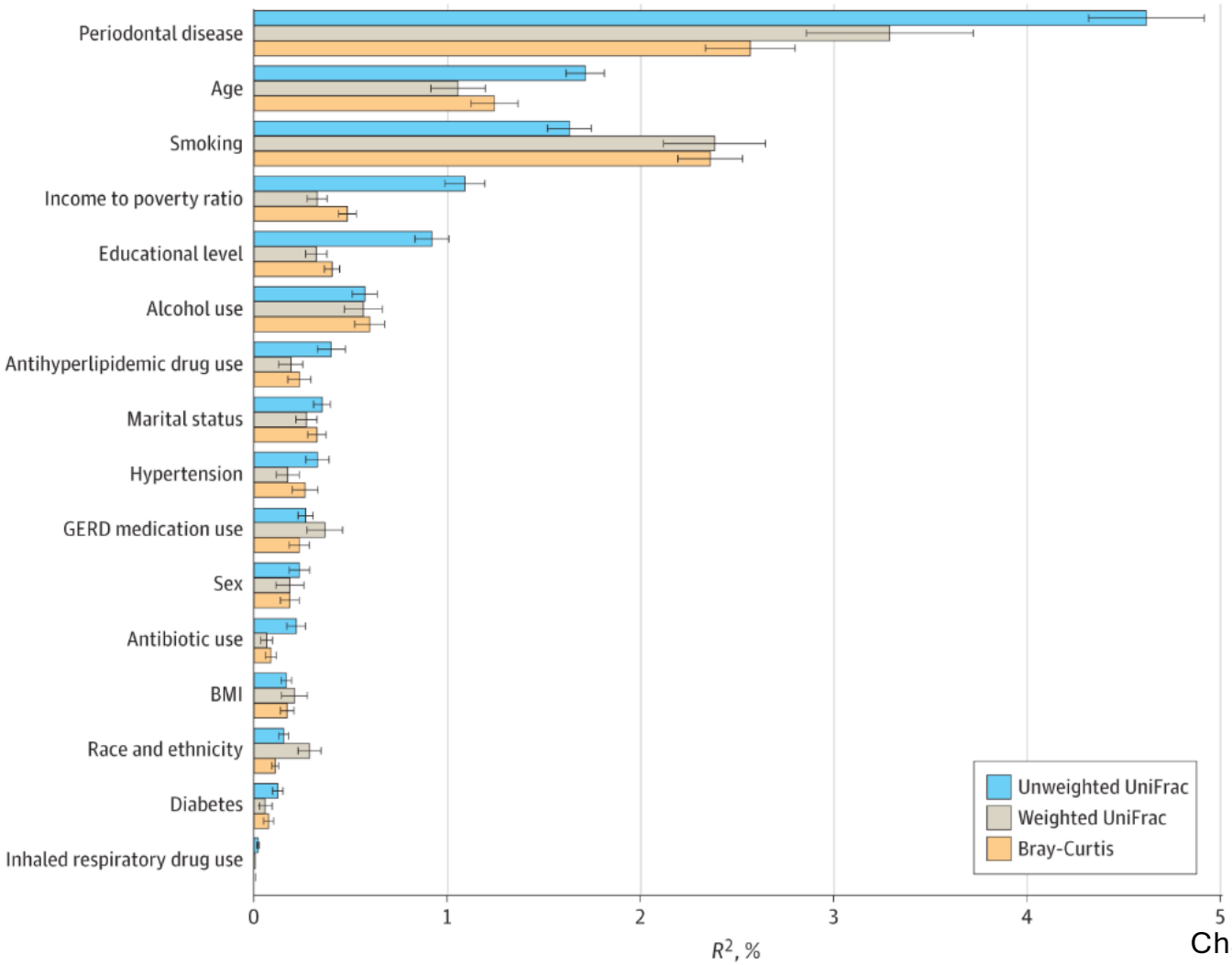
Statistical Microbiome Analysis with Survey Samples

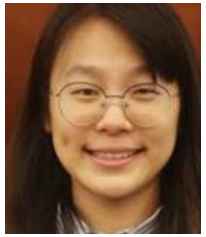
- **NHANES oral microbiome study**
  - Each sample has an inclusion probability. Weighted analysis to recover US average. Survey package in R or SAS.
- **Case-cohort oral microbiome study**
  - Controls selected to match with lung cancer cases by smoking, age, and sex. Cases have weight 1, controls have variable weights.
  - Weighted analysis to estimate unbiased HR

- **Beta-diversity analysis**
  - Weights & large sample sizes
  - *fast.adonis*
  - Efficient for many covariates simultaneously or incrementally



Chaturvedi et al, 2025; Vogtmann et al,2022, Li et al., 2022

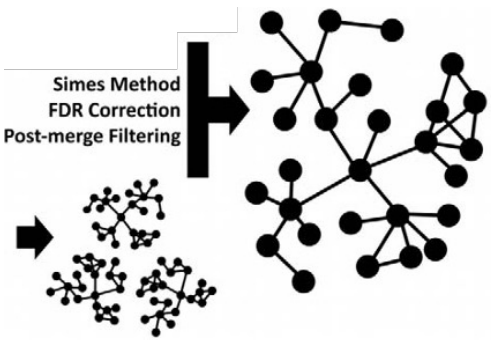




Chaturvedi et al, 2025; Li et al., 2022

Microbiome Network Analysis: Marginal vs. Partial Correlation

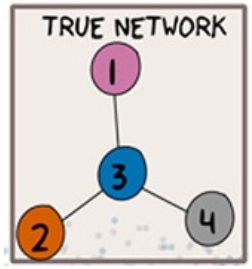
- **Complex ecosystem where microbes compete and cooperate**
- **Marginal correlation analysis**
  - Taxa are connected if marginally associated.
  - Pearson, Spearman, KL-distance …
  - Maybe indirect interaction.



- **Gaussian graphic model: partial correlation analysis**
  - Estimate partial covariance matrix with sparseness
  - Node-wide penalized regression analysis
  - *SPIEC-EAS*

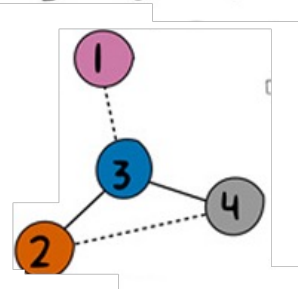
Kurtz et al, 2015; Faust et al,2012

Microbiome Network Analysis: Marginal vs. Partial Correlation



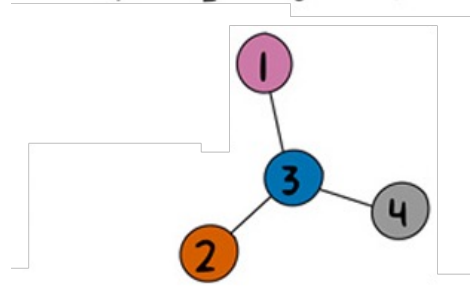
Marginal Correlation

	1	2	3	4
1	1	0.25	0.39	-0.3
2	0.25	1	0.64	-0.49
3	0.39	0.64	1	-0.76
4	-0.3	-0.49	-0.76	1



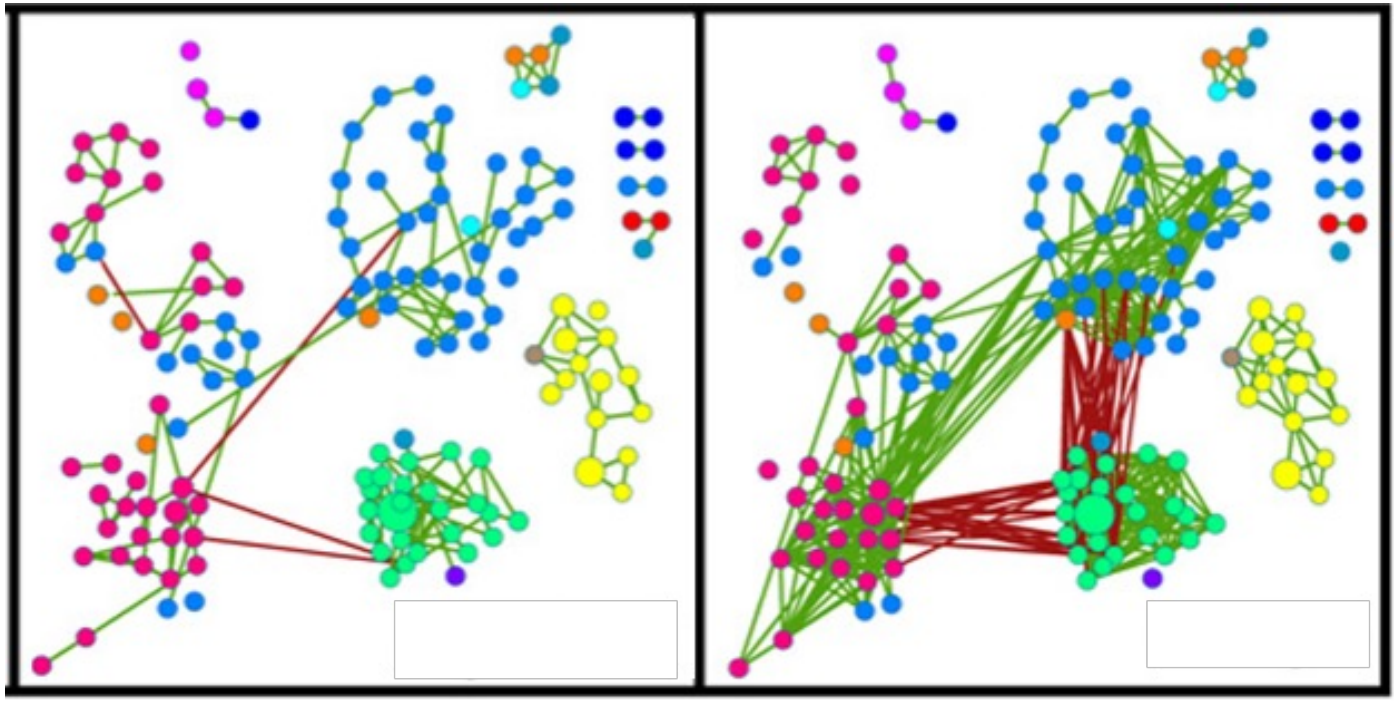
Partial Correlation

	1	2	3	4
1	1.2	~0	-0.47	~0
2	~0	1.7	-1.1	~0
3	-0.47	-1.1	3.3	1.8
4	~0	~0	1.8	2.4



Microbiome Network Analysis: Marginal vs. Partial Correlation

Partial correlation analysis **Marginal correlation analysis**



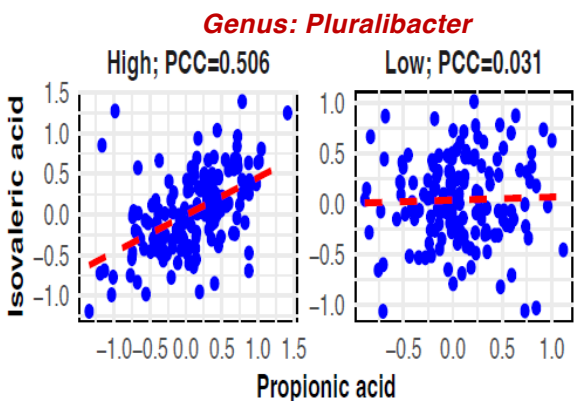
Green: positive; Red: negative. Node: OTU

SPIEC-EAS

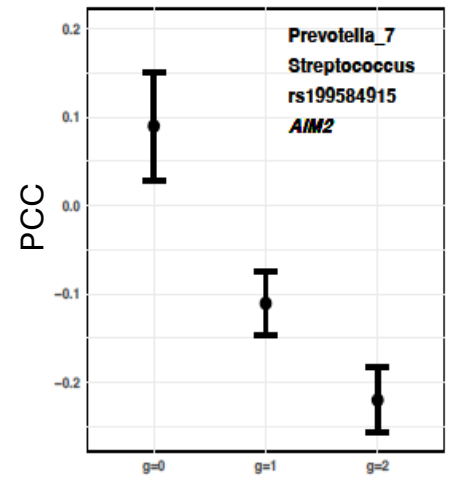
Kurtz et al, 2015

Network Analysis in Multi-Omics:
GFBioNet Identifies Genomic Factors Regulating Network

- **Microbiome, host genetics, metabolomics**
- **First order analysis**
  - Host genetic → microbiome taxa abundances, metabolome levels
  - Microbiome taxa → metabolome levels
- **Second order analysis**
  - Microbiome taxa → metabolome interactions.
  - Host genetics → taxa-taxa interactions



PLCO oral microbiome 1356 subjects
Prevotella_7* and *Streptococcus
Regulated by seven SNPs in immune related genes



Anyaso-Samuel et al, submitted;

NCI/DCEG Microbiome Working Group: Moving Microbiome Research into Population Studies

Since 2010, perform high-quality population studies to better understand cancer etiology, to improve risk prediction, and prevention of cancers.